

PROBLEM 15.21 Working drawing (in.)

Assembly Name: Worm Gear Reducer

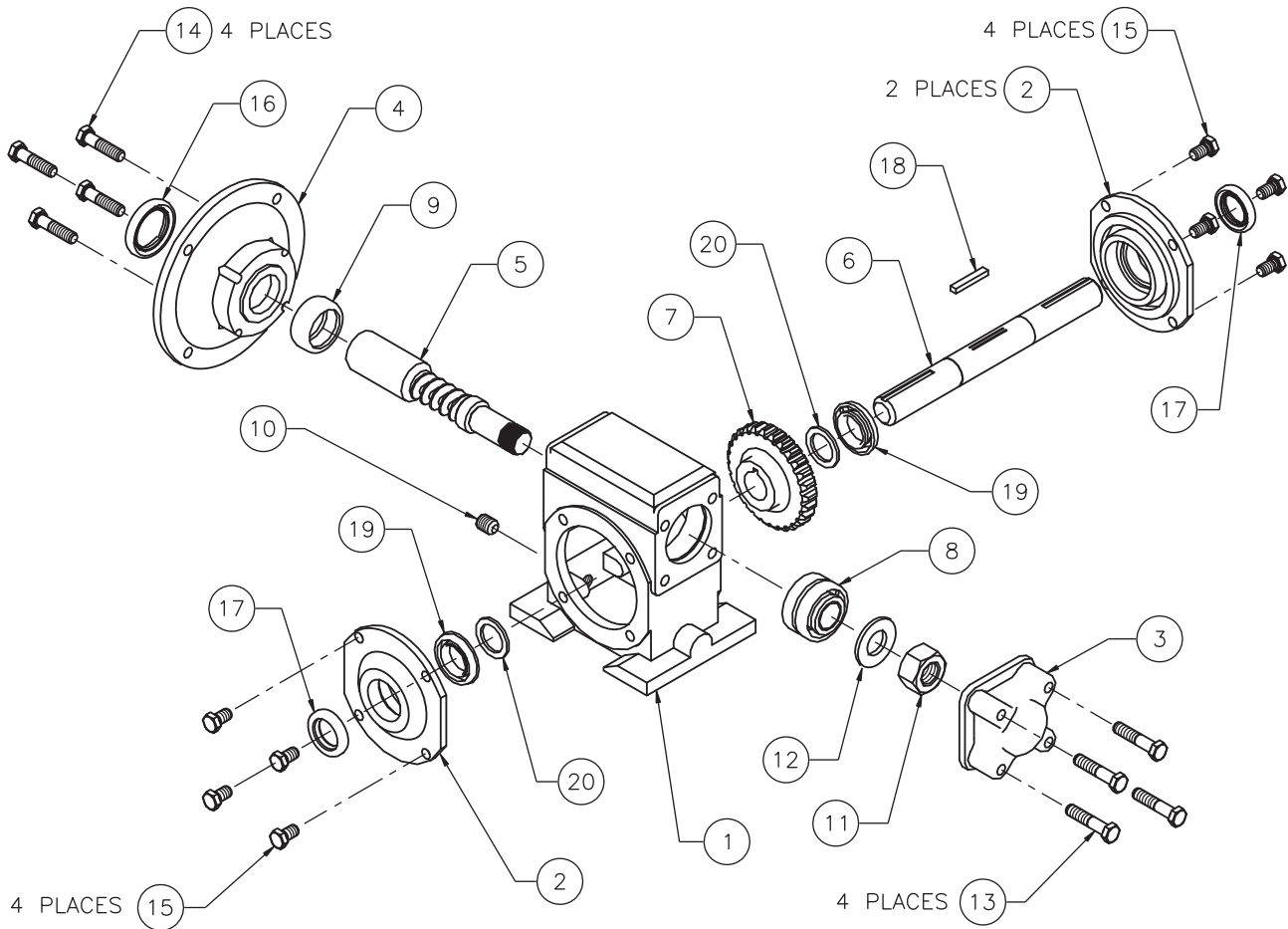
SPECIFIC INSTRUCTIONS: This problem is demanding, and it contains challenging applications that include gears and bearings. You should study gears and related drafting practices covered in Chapter 16 before completing this problem. This is an advanced project, and design changes may be required. Prepare a complete set of working drawings with one

detail drawing per sheet and the assembly and parts list on another sheet. When preparing the assembly drawing, use separate balloons for each part in each view or use only one balloon in the view that most clearly identifies the part. This is an engineering prototype project. Errors are likely in this problem. Verify dimensions during assembly. Establish tolerances between mating parts. A completed assembly drawing and parts list for this problem is shown in Figure 15.24.

PARTS LIST

ITEM	QTY	NAME	DESCRIPTION
1	1	HOUSING	
2	2	RETAINING PLATE	
3	1	BEARING CAP	
4	1	MOTOR ADAPTER	
5	1	HIGH SPEED SHAFT	
6	1	SLOW SPEED SHAFT	
7	1	WORM GEAR	BRONZE
8	1	DBL. ROW TAPERED ROLLER BEARING	KOYO46T30305DJ/29.5
9	1	SNGL. ROW CYL. ROLLER BEARING	KOYO CRL11
10	1	TAPER PLUG	500-16NPT PLUG
11	1	HEX NUT	.875-16 UN-28
12	4	HIGH SPEED LOCKWASHER	TIMKEN TW-105

ITEM	QTY	NAME	DESCRIPTION
13	4	MACHINE SCREW	.375-16UNC-2A × 1.813 HEX HEAD
14	4	MACHINE SCREW	.375-16UNC-2A × 1.625 HEX HEAD
15	8	MACHINE SCREW	.375-16UNC-2A × .625 HEX HEAD
16	1	HIGH SPEED OIL SEAL	PARKER 2-028
17	2	SLOW SPEED OIL SEAL	PARKER 2-020
18	1	SLOW SPEED KEYWAY	.1875 × .245 × 1.450
19	1	SNGL. ROW TAP ROLLER BEARING	KOYO 32005J
20	2	SLOW SPEED SPACER	TIMKEN TW-506

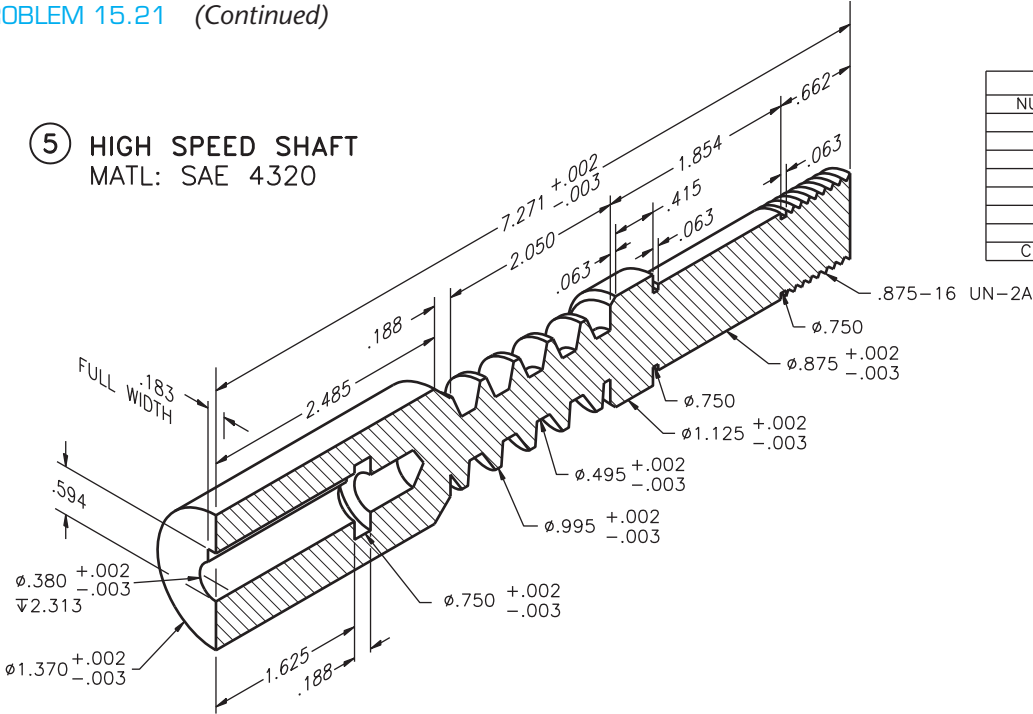






PROBLEM 15.21 (Continued)

5 HIGH SPEED SHAFT
MATL: SAE 4320



WORM GEAR DATA	
NUMBER OF THREADS	5
AXIAL PITCH	
PRESSURE ANGLE	20°
PITCH DIAMETER	.750
LEAD RIGHT HAND	.300
LEAD ANGLE	169°
ADDENDUM	.125
WHOLE DEPTH	.250
CHORDAL THICKNESS	.163

7 WORM GEAR
MATL: PHOSPHOR BRONZE

SPUR GEAR DATA	
DIAMETRAL PITCH	8
NUMBER OF TEETH	27
PRESSURE ANGLE	20°
PITCH DIAMETER	3.375
BASE CIRCLE DIAMETER	3.6187
CIRCULAR PITCH	.3927
CIRCULAR THICKNESS	.1964
ROOT DIAMETER	3.125

